



Lesson 2.3: Software

https://codehs.com/lms/assignment/93877822/lesson_plan

Description	What kinds of software do computers use and need? In this lesson, the topic of software is broken down into types of software, how they interact, and the specific functions of the different types of software.								
Objective	Students will be able to: • Understand and identify different types of software and their functions								
Activities	2.3.1 Video: Software 2.3.2 Check for Understanding: Software Quiz 2.3.3 Connection: Software Explained 2.3.4 Free Response: Computer Applications You Use 2.3.5 Free Response: Operating Systems								
Prior Knowledge	• basic components of a computer system and how they are organized								
Planning Notes	• Review the slides and the exercises in the lesson. • Collect images of older operating systems such as Windows 3.0 and MS-DOS. There are emulators available as well to really show students older operating systems. • This is a shorter lesson and can be paired with the next lesson.								
Standards Addressed	CompTIA Tech Standards <table><tr><th>Name</th><th>Description</th></tr><tr><td>3.3</td><td>Explain the purpose and proper use of software.</td></tr></table> CSTA 3A Standards <table><tr><th>Name</th><th>Description</th></tr><tr><td>3A-CS-02</td><td>Compare levels of abstraction and interactions between</td></tr></table>	Name	Description	3.3	Explain the purpose and proper use of software.	Name	Description	3A-CS-02	Compare levels of abstraction and interactions between
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application software, system software, and hardware layers.

England 3 Standards

Name	Description
ENG.3.5	understand the hardware and software components that make up computer systems, and how they communicate with one another and with other systems

PRAXIS 5652 Standards

Name	Description
V.A.1	Know that operating systems are programs that control and coordinate interactions between hardware and software components
V.A.1.b	identify software components and their functions
V.A.1.c	identify common operating systems tasks
V.A.4	Be familiar with computers as layers of abstraction from hardware (e.g., logic gates, chips) to software (e.g., system software, applications)
V.A.4.a	identify appropriate abstraction layers for hardware and software components

Teaching and Learning Strategies

Lesson Opener:

- Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins]

Activities:

- Watch the lesson video and take the corresponding quiz. This quiz is a quick check for understanding. [10 mins]
- Explore the *Software Explained* connection. [5 mins]

	◦ Have students take notes on the differences of user application software and operating system software. • Complete the <i>Computer Applications</i> free responses activity. [5 mins] ◦ Choose one of the five software applications and research when it was created, who created it, and how has it been improved since it was first created. • Complete the <i>Operating Systems</i> reflection. [5 mins] ◦ Have students explore older versions of Windows. What was there BEFORE the Windows operation system? Lesson Closer: • Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]
Discussion Questions	Beginning of class: • You've probably heard the term operating system. What do you think this is? ◦ <i>Answers will vary.</i> • Do you know what operating system your computer is using? ◦ <i>Answers will vary.</i> • Why do you think your computer needs an operating system? ◦ <i>Answers will vary.</i> End of class: • What are a few examples of user application software? ◦ <i>Games, office applications, word processors, spreadsheet applications, web browsers, etc.</i> • What is the difference between user application software and system software? ◦ <i>User application software are programs that run on the computer that help users perform tasks. System software are programs that help the computer perform tasks.</i> • What does a computer operating system do? ◦ <i>In short, it makes your computer usable by acting as a layer between user applications and hardware.</i>
Resources/Handouts	

Vocabulary

Term	Definition
Computer	A person or device that makes calculations, stores data, and executes instructions according to a program.
Software	Programs that can be run on a computer

Modification: Advanced	Modification: Students Needing Additional Support	Modification: English Language Learners
	• Print out video slides for students to reference	• Print out video slides for students to reference